ArogyaX — AI medical chatbot

Institution
Bennett University, B.Tech CSE (AIML)

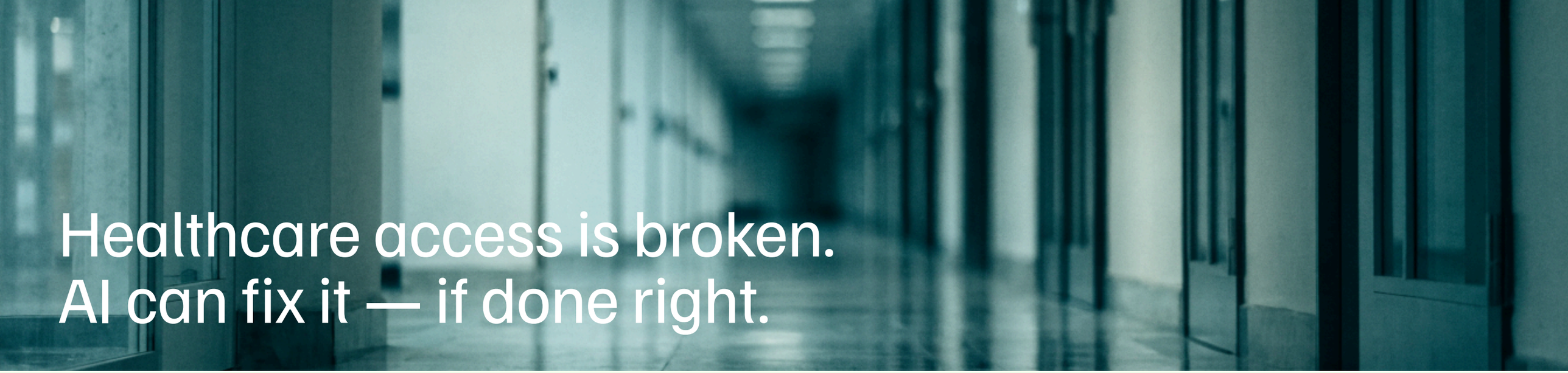
Supervised by
Dr. Prashant Shukla

Team
Ayush Upadhyay
Riti Dubey
Gaurav
Tanistha Keshri

Abstract

Healthcare is changing fast, and AI is at the center of that shift. ArogyaX is an AI-powered medical chatbot built to make health information more accessible and easier to understand.
Research problem: Traditional healthcare systems struggle with availability gaps, long wait times, and the inability to interpret unstructured medical documents at scale.

Objectives and methodology: The system uses NLP and Retrieval-Augmented Generation (RAG) to understand queries and respond with context-aware answers. OCR and image processing handle PDFs, handwritten prescriptions, and X-ray images.
Key findings: Testing showed ArogyaX reliably answers medical questions, interprets reports, and provides fast, accessible first-line health assistance — while making clear it does not replace professional medical advice.



Healthcare access is broken. AI can fix it — if done right.

01

Remote access gaps

Specialists are hard to reach in rural and underserved areas, leaving patients without timely medical guidance.

02

Long wait times

Patients lose critical time waiting for consultations that could be addressed instantly with an intelligent system.

03

Unreadable reports

Most patients cannot interpret their own lab results, prescriptions, or medical documents without expert help.

04

Rule-based chatbots fall short

Existing bots lack multimodal understanding — they cannot process images, scans, or handwritten notes.

01 Input and extraction

User submits text, PDF, scanned note, or medical image. NLP and OCR extract structured meaning from all input types.

02 Preprocessing and embeddings

Data is cleaned, preprocessed, and converted into vector embeddings using Hugging Face Transformers for semantic representation.

03 Knowledge retrieval

FAISS or ChromaDB retrieves the most relevant context from a curated medical knowledge base using similarity search.

04 LLM response generation

Llama 3 via Groq generates a grounded, context-aware response, displayed in a clean FastAPI-powered interface.

Literature review

AI-powered chatbots for medicine

Early medical chatbots were rule-based — limited in handling complex queries. Recent systems use Machine Learning and Deep Learning to understand user intent and respond contextually, supporting symptom assessment, mental health, and general medical queries.

NLP in healthcare

NLP extracts meaning from patient queries and medical literature. Transformer-based models lead the field, understanding words in context — critical for accurate medical terminology handling.

Retrieval-Augmented Generation (RAG)

RAG combines retrieval and generation. Rather than relying solely on model knowledge, it searches a curated knowledge base first, then generates answers from retrieved content. This reduces hallucinations — a major concern in medical AI — making RAG the most reliable architecture for medical chatbots today.

Literature review

OCR in healthcare

OCR converts scanned or handwritten text into machine-readable format — used widely for prescriptions, lab results, and records. TrOCR (transformer-based) has significantly improved handwriting recognition, making it viable for medical document intake.

Multimodal AI systems

Multimodal AI processes text, images, and documents together. Research confirms combining modalities improves decision quality, though integration of specialised models remains technically complex.

Gaps in existing systems

Most medical chatbots fall short in critical ways:

- Heavy reliance on fixed datasets with low adaptability
- No support for images or handwritten inputs
- Risk of context-independent or hallucinated responses
- No conversation memory or personalisation

These gaps motivated ArogyaX — a flexible, multimodal, RAG-powered assistant.

Data collection
Supported input types

Text queries
Users ask health questions in natural language. Example: "What does high creatinine mean?"

PDF reports
Blood tests, diagnoses, and clinical summaries — typed or handwritten — are accepted and parsed.

Prescriptions
Doctor's notes are extracted using TrOCR, a model specialised in handwritten text recognition.

Medical images
X-rays and kidney scans are accepted and pre-processed using OpenCV and PIL.

Synthetic data
Artificial patient reports with demographics, diagnoses, and lab results used for benchmarking.

Preprocessing
All inputs undergo noise removal, OCR extraction, resizing, and embedding into vector format.

Tools and technologies
ArogyaX AI stack

Core AI and NLP
LangChain · Hugging Face · Groq API (Llama 3.3-70b) · PyTorch

Document processing
PyMuPDF · pdfplumber · Tesseract OCR · TrOCR (handwriting)

Image processing
OpenCV for pre-processing · PIL for image loading and formatting

Retrieval and storage
ChromaDB for vector embeddings · FAISS for fast similarity search

Backend
FastAPI bridges the Python AI backend with the frontend interface

Design principle
Modular, open-source stack — each layer independently upgradeable

RAG knowledge base

The chatbot queries a curated offline vector database built from 11 verified medical textbooks — no internet search. Every response is grounded in clinical literature.

- Netter's Atlas of Human Anatomy
- Braunwald's Heart Disease
- Davidson's Principles and Practice
- Goodman & Gilman's Pharmacology
- Gray's Anatomy for Students
- Guyton Medical Physiology
- Harrison's Internal Medicine, 21st Ed.
- Nelson Textbook of Pediatrics
- Oxford Handbook of Oncology
- Williams Obstetrics, 24th Ed.

Chunking statistics
Total text chunks: 149,398
Characters indexed: 65,880,275
Average chunk size: 441 characters
Chunk overlap: 50 characters
Embedding model
sentence-transformers/all-MiniLM-L6-v2
Vector dimensions: 384 per chunk

End-to-end workflow

01 Input and extraction

Text, PDF, image, or handwritten input. OCR (Tesseract / TrOCR) and PyMuPDF extract raw content.

02 Preprocess and embed

Text cleaned and normalised. Converted to a 384-dimensional vector using all-MiniLM-L6-v2.

03 Retrieve and generate

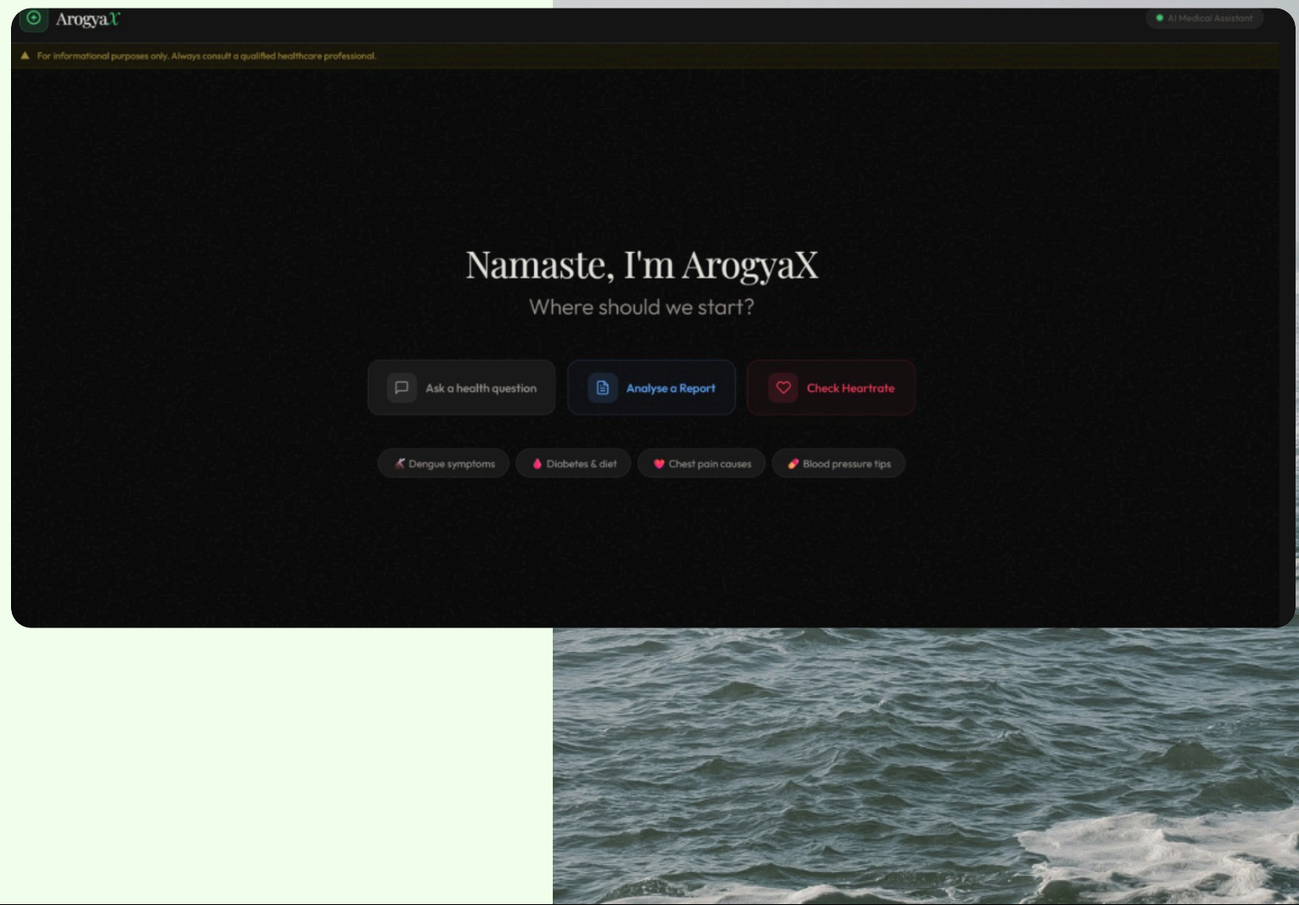
FAISS / ChromaDB similarity search with MMR selects 5 chunks. Llama 3 (Groq) generates a grounded answer.

04 Output and rPPG

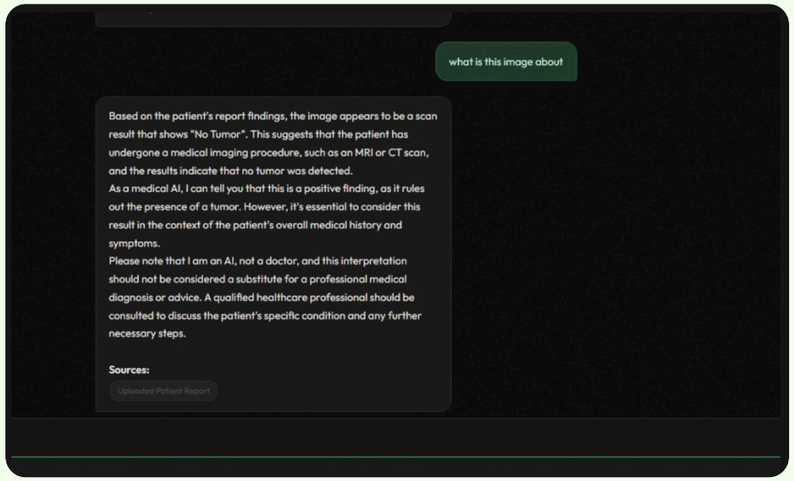
Response rendered in Markdown with source citations. Webcam rPPG module extracts heart rate via FFT on green channel.

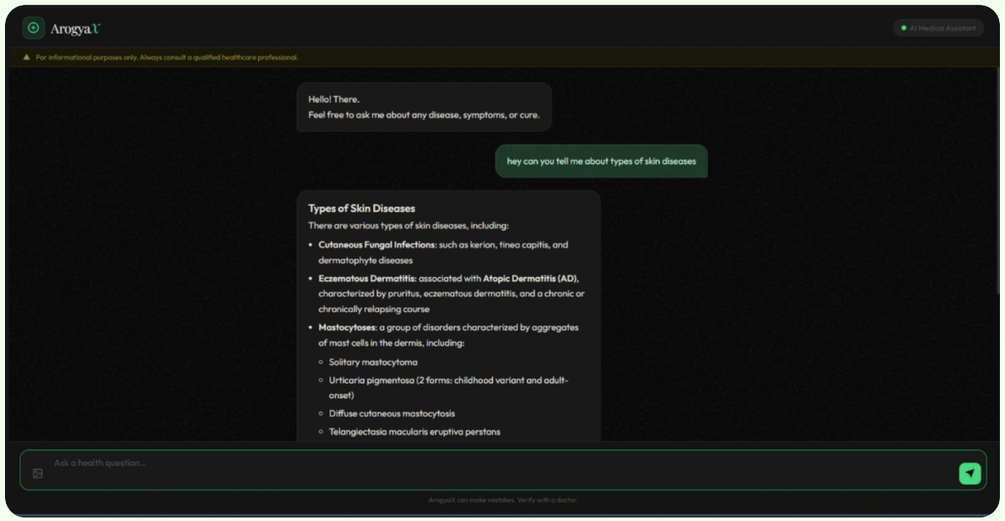
Your AI health companion, live.

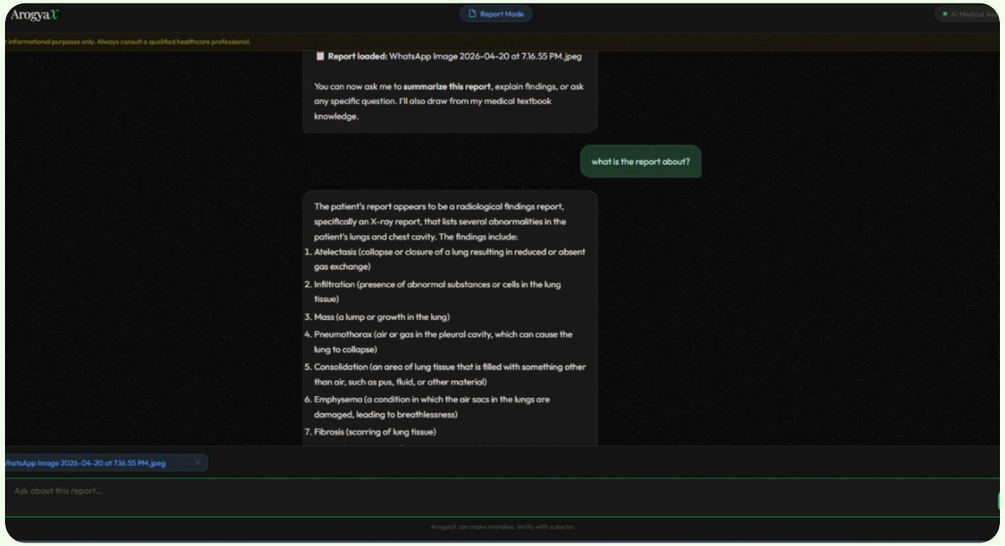
Ask a health question, analyse a report, or check heart rate — ArogyaX routes every query to the right pipeline automatically, with clinical sources and a care disclaimer.



Report analysis and image diagnosis







Report analysis

- 7 chest X-ray abnormalities identified
- Atelectasis, infiltration, mass
- Pneumothorax, consolidation
- Emphysema, fibrosis
- Symptom queries grounded in textbooks

Image diagnosis

- Brain MRI/CT scan analysed
- "No tumor detected" returned
- Scan type identified and noted
- Clinical interpretation included
- Disclaimer: consult a professional

Performance analysis

Key observations

ArogyaX handles general health and symptom-based queries effectively. Clear, specific queries yield more accurate responses than broad or highly specialised clinical questions.

Text inputs are faster than image or PDF inputs.

Performance remained consistent across all test runs — no degradation detected.

Verification methodology

A diverse set of user queries was tested. Chatbot responses were compared against established medical reference data — all evaluated responses were relevant, appropriate, and contextually accurate.

Testing was conducted under varied conditions to confirm robustness across edge cases.

Key findings

3-in-1

Input types: text, PDF, images

149,398

Text chunks indexed from 15 medical textbook

24/7

AI health guidance without doctor presence

384-dim

Embeddings powering real-time symptom triage

Info only

Clinical decisions require qualified professionals

Conclusion

ArogyaX demonstrates that AI can be practically applied to healthcare assistance. By integrating NLP, RAG, OCR, and image processing into a single modular system, the platform understands diverse inputs and generates clinically grounded responses.

The retrieval-based approach is a key strength: rather than relying on model memory alone, ArogyaX searches verified medical textbooks before answering, reducing hallucinations and improving reliability.

Testing confirmed the system is effective, consistent, and accessible — while clearly communicating its limitations. It is a tool for awareness and preliminary guidance, not a substitute for medical professionals.

Future work

- Integrate specialised medical AI models for higher accuracy
- Improve TrOCR on complex handwritten prescriptions
- Expand image analysis to broader diagnostic imaging
- Add a voice interface for hands-free interaction
- Connect to live medical databases and real-time APIs
- Build user history and personalisation features
- Deploy ArogyaX as a mobile application

References

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